Maths Teacher, 5A

Vasant Valley School, New Delhi

Subject: Maths

3139/M5A/0002

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Maths Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Maths are:

Applications in daily life: word/visual problems
Number sense, related concepts and basic number competency

Great job nurturing these talents!

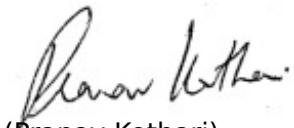
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Maths :

Problem solving: advanced or challenging problems
Arithmetic operations: four basic operations, properties

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	C	C	D	C	C	B	C	B	D	C	B	D	A	C	C	B	D	D	C	C	D	D	A	B	A	D	A	C	A	C	D	D	D	D	B	A	C	A	D	D				
1	Adhrit P Chandra		✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	D	B	✓	✓	✓	34	682	94	
2	Akira Nagpal		✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	A	A	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	C	A	C	C	C	D	✓	D	✓	C	27	597	80	
3	Annika I Dayal		✓	✓	B	A	✓	✓	✓	D	✓	✓	✓	✓	✓	A	✓	A	C	A	✓	✓	C	✓	✓	C	✓	✓	✓	A	✓	B	✓	A	✓	✓	D	C	B	✓	✓	C	24	568	73	
4	Aranya D Bedi		✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	A	A	D	✓	✓	✓	✓	D		✓	✓	✓	D	✓	B	✓	C	D	✓	✓	✓	✓	✓	29	613	83	
5	Arav Saha		✓	✓	✓	A	✓	✓	✓	D	A	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	A	✓	C	✓	✓	✓	✓	✓	D	✓	✓	B	✓	D	✓	✓	D	✓	A	29	611	83	
6	Archisha Singh		✓	B	C	✓	✓	✓	*	*	✓	A	D	✓	B	B	A	D	✓	A	A	✓	C	✓	✓	✓	✓	✓	C	✓	✓	B	B	C	✓	✓	C	C	*	✓	✓	✓	20	515	58	
7	Asta A Dayal		✓	A	B	✓	A	✓	✓	C	✓	B	D	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	B	✓	✓	✓	✓	A	✓	✓	✓	✓	A	✓	C	✓	27	580	76	
8	Ayaan Malik		D	D	A	B	D	D	A	✓	C	D	D	B	C	✓	B	C	✓	A	B	A	A	✓	✓	D	C	B	D	B	C	A	A	A	✓	B	✓	C	✓	D	C	B	8	375	5	
9	Ayaan Tandon		✓	A	C	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	34	700	95	
10	Ayat Rehman		✓	A	✓	✓	✓	✓	A	✓	✓	A	✓	✓	✓	✓	A	✓	B	C	✓	D	A	A	✓	✓	✓	B	B	✓	✓	B	✓	A	✓	C	✓	C	D	✓	✓	A	23	546	68	
11	Bharti		✓	✓	A	✓	✓	C	B	C	A	A	A	✓	C	B	D	C	✓	✓	B	B	A	A	D	A	C	C	B	B	✓	A	✓	B	C	✓	✓	C	A	C	C	✓	12	419	18	
12	Dipanshi		✓	✓	C	✓	✓	✓	✓	C	✓	A	D	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	A	C	✓	✓	✓	✓	30	627	86	
13	Gaia Bhagwani		✓	*	✓	✓	A	✓	D	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	35	688	94	
14	Karan Puri		✓	✓	C	✓	✓	✓	✓	D	A	✓	✓	✓	✓	D	B	✓	✓	✓		✓	A	A	*	D	C	✓			✓	B	C	✓		✓							18	501	52	
15	Mohammad Faizan		✓	✓	C	✓	✓	✓	✓	C	✓	✓	D	✓	✓	B	✓	A	✓	A	A	✓	C	✓	C	✓	✓	C	B	✓	✓	✓	C	C	B	✓	D	B	D	✓	✓	✓	23	532	63	
16	Nabiya Chouhan		B	A	C	D	B	C	A	✓	C	D	D	C	✓	B	A	✓	✓	A	✓	B	A	✓	B	C	B	A	B	D	C	A	✓	B	A	B	C	C	A	D	B	C	7	350	1	
17	Neel R Khanna		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	C	✓	✓	✓	C	✓	A	✓	✓	A	✓	D	✓	✓	✓	✓	✓	D	C	A	✓	C	✓	C	B	C	✓	✓	27	589	79	
18	Parth H Yadav		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	37	748	98	
19	Prathit Goyal		A	✓	✓	B	✓	C	A	A	B	✓	D	B	C	B	A	C	A	C	*	B	A	B	C	D	C	*	B	A	C	A	A	B	C	A	✓	*	D	B	A	C	5	323	0	
20	Reyansh Chopra		✓	A	✓	✓	✓	✓	A	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	B	✓	✓	D	✓	✓	B	D	✓	B	✓	✓	B	*	✓	✓	✓	B	B	C	26	575	75	
21	Ruhani Tulshan		B	✓	✓	✓	D	✓	✓	C	✓	D	✓	✓	✓	B	✓	✓	✓	✓	✓	B	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	C	✓	✓	C	B	✓	✓	C	28	593	79	
22	Sameerudin Ansari		✓	A	✓	D	B	✓	✓	✓	A	A	D	C	D	A	B	D	C	B	A	✓	C	A	B	D	C	A	B	D	✓	✓	C	✓	B	A	D	✓	B	C	✓	B	11	416	17	
23	Samerah J Srinivas		✓	B	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	A	A	✓	✓	✓	✓	✓	C	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	33	666	92	
24	Sanshray J Pratyush		✓	✓	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	D	✓	✓	✓	✓	✓	D	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	35	742	98	
25	Shivai Dang		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	D	✓	✓	✓	A	✓	C	✓	✓	✓	✓	✓	✓	✓	35	714	96	
26	Shreya D Bahree		✓	B	C	✓	✓	✓	✓	D	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	C	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	32	655	90	
27	Sofia C Sanjanwala		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	B	✓	B	✓	✓	✓	C	✓	✓	✓	✓	36	747	98	
28	Taahira Trehan		✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	C	B	✓	✓	✓	✓	✓	✓	✓	D	C	✓	✓	✓	*	C	A	B	C	D	C	B	✓	C	C	25	566	73	
29	Vaidehi Chaudhry		✓	✓	✓	✓	✓	✓	✓	D	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	37	756	99	O
30	Vieara Chaudhry		✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	B	✓	B	C	B	✓	✓	✓	C	B	D	✓	C	30	618	84	

% answered correct (section)	82	68	62	79	79	88	71	35	76	59	62	82	82	65	65	71	76	56	56	71	41	71	79	53	76	68	59	65	85	32	59	41	59	62	65	44	50	59	74	59	585.8/500			
% answered correct (school students)	80	79	63	72	73	87	68	37	73	58	61	74	76	71	67	56	68	48	51	54	54	60	69	60	67	78	62	69	79	38	56	39	66	63	65	45	49	58	65	54	579.1/500			
% answered correct (all students)	60	66	49	51	59	79	56	36	50	35	47	64	62	52	62	44	58	36	45	49	48	44	54	34	43	67	48	44	66	40	45	29	53	59	54	35	35	47	52	45	506.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**C**	**C**	**D**	**C**	**C**	**B**	**C**	**B**	**D**	**C**	**B**	**D**	**A**	**C**	**C**	**B**	**D**	**D**	**C**	**C**	**D**	**D**	**A**	**B**	**A**	**D**	**A**	**C**	**A**	**C**	**D**	**D**	**D**	**D**	**B**	**A**	**C**	**A**	**D**	**D**				
31	Yadav Anushka		✓	✓	✓	✓	✓	✓	✓	D	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	D	✓	✓	✓	✓	✓	D	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	34	695	95	
32	Yash Yadav		B	✓	✓	✓	B	✓	*	A	A	A	D	B	✓	B	✓	✓	✓	C	B	✓	A	✓	B	✓	✓	C	✓	B	C	B	C	A	B	A	✓	✓	B	D	A	C	15	436	25	
33	Yuvaan Bhutani		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	A	✓	✓	✓	✓	✓	✓	✓	✓	A	C	B	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	34	667	92	
34	Zayn Singh		B	✓	A	✓	✓	✓	A	C	✓	D	D	✓	✓	B	B	✓	B	C	A	✓	A	✓	✓	D	✓	✓	D	D	✓	✓	✓	C	B	B	✓	C	B	D	✓	✓	19	506	54	

% answered correct (section)	82	68	62	79	79	88	71	35	76	59	62	82	82	65	65	71	76	56	56	71	41	71	79	53	76	68	59	65	85	32	59	41	59	62	65	44	50	59	74	59	585.8/500			
% answered correct (school students)	80	79	63	72	73	87	68	37	73	58	61	74	76	71	67	56	68	48	51	54	54	60	69	60	67	78	62	69	79	38	56	39	66	63	65	45	49	58	65	54	579.1/500			
% answered correct (all students)	60	66	49	51	59	79	56	36	50	35	47	64	62	52	62	44	58	36	45	49	48	44	54	34	43	67	48	44	66	40	45	29	53	59	54	35	35	47	52	45	506.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 5 A, has performed better than the national average in the Maths ASSET paper this year.

Class 5	Maths	No. of students
5A (Your Class)	117%	34
5B	112%	32
5C	118%	32
All sections combined	116%	98

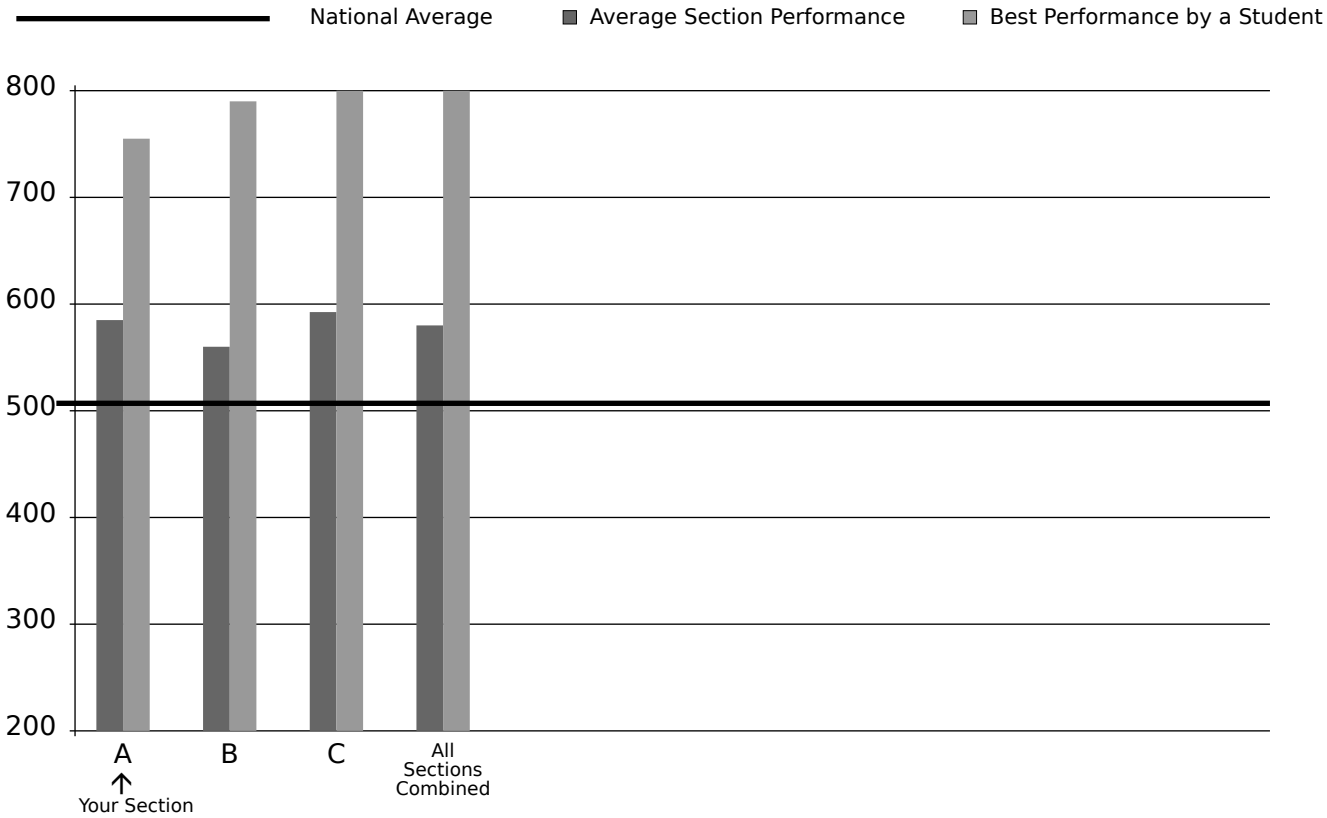
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

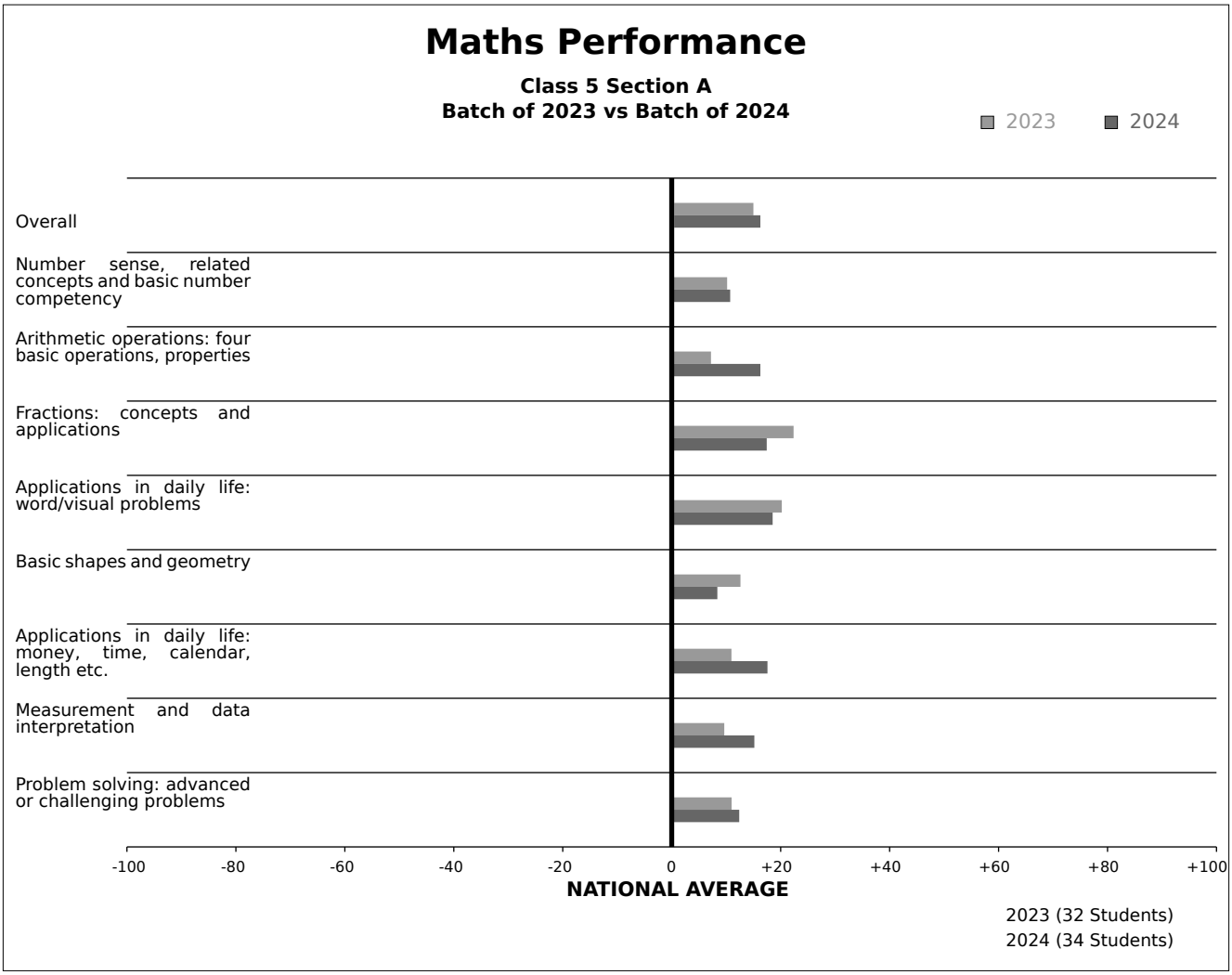
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 5 A Subject: Maths Number of Students: 34

S.No.	Skill	Questions	Class 5 A Performance	Class 5 Performance (School)	Class 5 Performance (National)
			Average	Average	Average
1*	Number sense, related concepts and basic number competency	1,14,21,23,26	67.1%	70.4%	56.3%
2*	Arithmetic operations: four basic operations, properties	5,28,31,32,40	60.6%	58.4%	44.3%
3*	Fractions: concepts and applications	11,13,16,27,37	64.7%	60.8%	47.3%
4*	Applications in daily life: word/visual problems	4,6,7,12,20	78.2%	71.2%	59.7%
5*	Basic shapes and geometry	2,9,17,30,34	62.9%	64.3%	54.6%
6*	Applications in daily life: money, time, calendar, length etc.	3,8,10,25,29	63.5%	60.8%	46.0%
7*	Measurement and data interpretation	19,22,33,35,39	64.7%	61.6%	49.5%
8*	Problem solving: advanced or challenging problems	15,18,24,36,38	55.3%	55.7%	42.9%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Applications in daily life: word/visual problems	1.	Problem solving: advanced or challenging problems
2.	Number sense, related concepts and basic number competency	2.	Arithmetic operations: four basic operations, properties

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 30

How many square tiles (as shown above) will be needed to cover the rectangular area below?

A 4 **B** 6 **C** 8 **D** 10

Skill: Basic shapes and geometry

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
30	25224	C	32%	40%	Your School / All Schools

Q 21 97 + 98 + 99 + 100 + 101 + 102 + 103 = 700.

Which of these is the same as 98 + 99 + 100 + 101 + 102?

(Note: You need not calculate)

A 700 - 97 **B** 700 - 100 **C** 700 - 103 **D** 700 - 200

Skill: Number sense, related concepts and basic number competency

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
21	25224	D	41%	48%	Your School / All Schools

Q 8 Which of these figures shows a water melon weighing exactly 2 kg?

Skill: Applications in daily life: money, time, calendar, length etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
8	25224	B	35%	36%	Your School	
					All Schools	

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 25 Sudhir took a flight from Mumbai to Delhi at 7:45 PM. He reached Delhi at 9:15 PM.

How long did the flight take to reach Delhi?

A 1 hour 30 minutes
B 2 hours
C 2 hours and 30 minutes
D 3 hours

Skill: Applications in daily life: money, time, calendar, length etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
25	25224	A	76%	43%	Your School	
					All Schools	

Q 4 Manya made candles and packed them in boxes such that each box had 6 candles. 2 candles remained after packing.

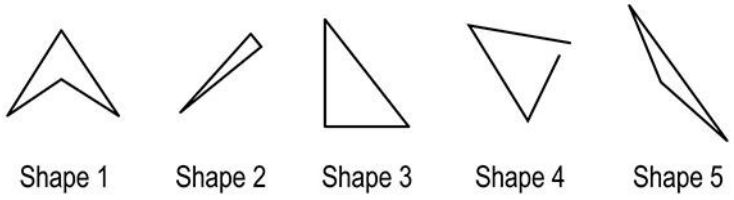
Which of the following could be the number of candles she made?

A 16
B 18
C 20
D 22

Skill: Applications in daily life: word/visual problems

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
4	25224	C	79%	51%	Your School	
					All Schools	

Q 9 Which of these is a triangle?



A only shape 3
B only shapes 3 and 4
C only shapes 1, 3 and 4
D only shapes 2, 3 and 5

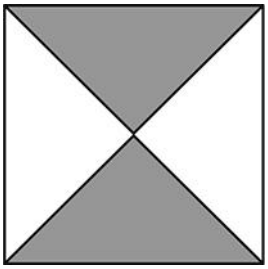
Skill: Basic shapes and geometry

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
9	25224	D	76%	50%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 11 What fraction of the figure below is shaded?

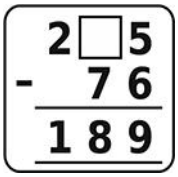


A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{2}$ **D** $\frac{4}{2}$

This question tests the ability to identify equivalent fractions. Option B is the correct answer as 2 out of 4 parts are shaded, which when simplified is 1/2. Option A shows the error of answering for the fraction of one shaded part instead of two. Option C shows an incorrect understanding of fraction as the number of parts shaded / the number of parts unshaded. Option D shows a flawed understanding of fraction as total number of parts/ number of parts shaded.

Papercode: 25224, Question no: 11					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	6%	62%	0%	32%	34
National	8%	47%	17%	28%	14049

Q 31 The subtraction problem shown below is solved correctly.



What digit goes into the empty box?

A 1 **B** 4 **C** 5 **D** 6

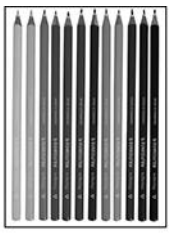
This question tests whether students have understood the subtraction-WITH-borrow algorithm. Students were expected TO identify the two occurrences of borrowing Ã”Ã‡Ã´ FROM the hundreds TO the tens place, AND FROM the tens TO the ones place. SOME students have only observed the former, thereby finding that TO subtract 7 tens AND get 8 tens IN the answer, the missing digit must have been 5 (FROM 15 tens). Draw studentsÃ”Ã‡Ã- attention TO the ones COLUMN AND the tens COLUMN Ã”Ã‡Ã´ HELP them see that borrowing has occurred IN BOTH places AND thus identify the missing digit AS 6 (7 tens have been subtracted FROM 15 tens, but 1 ten has already been borrowed IN ORDER TO be able TO subtract 6 ones FROM 15 ones). Students may also realise that adding 189 AND 76 would HELP them arrive AT the solution.

Papercode: 25224, Question no: 31					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	9%	3%	29%	59%	34
National	15%	10%	28%	45%	14049

Q 10 Archana bought TWO of the items shown below.

Rs 44

Rs 15



Rs 27

If she paid with a 100-rupee note, what is the MAXIMUM amount of change she could have received?

A Rs 29 **B** Rs 56 **C** Rs 58 **D** Rs 85

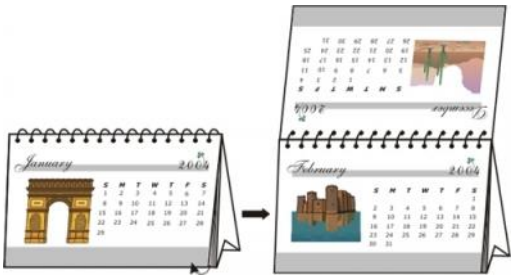
This question tests whether the student understands the relation between the bill value and the amount of change received. Students were expected to realise that in order to get the maximum amount of change, the bill value should be the least, and thus the two least expensive items should be purchased. Many students have associated the word Ã”Ã‡Â£maximumÃ”Ã‡Ã˜ with the maximum priced items and thus subtracted the sum of 44 and 27 from 100 to choose option A. Other students have understood the least expensive items should be purchased, but have missed the word Ã”Ã‡Â£twoÃ”Ã‡Ã˜ and subtracted only 15 from 100. Help students understand that the smaller the number subtracted, the larger the answer, and vice-versa. Allow them to observe this by engaging them in shopkeeper-customer role play in the classroom.

Papercode: 25224, Question no: 10					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	26%	3%	59%	12%	34
National	35%	9%	35%	18%	14049

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Problem solving: advanced or challenging problems (55%)

Q 1 A desk calendar concists of 6 sheets , with a different month on EACH side of the sheet.



The first 6 months appear in order as the sheets are turned over one by one. The months in the second half of the year are printed on the flip side of these sheets in order, but starting from BACK to FRONT. That is, December appears on the flip side of the first month, January.
Which month will be on the flip side of the page showing April?

A May **B** September **C** October **D** August

Explanation:

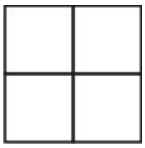
B: Let us try writing out the months in two rows. We will write the first 6 months in the first row, and in the second row we will write the months that appear on the flip side, in reverse order.

This is what we get.

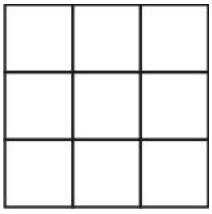
January	February	March	April	May	June
December	November	October	September	August	July

From the table it is clear that September will be on the flip side of April.

Q 2

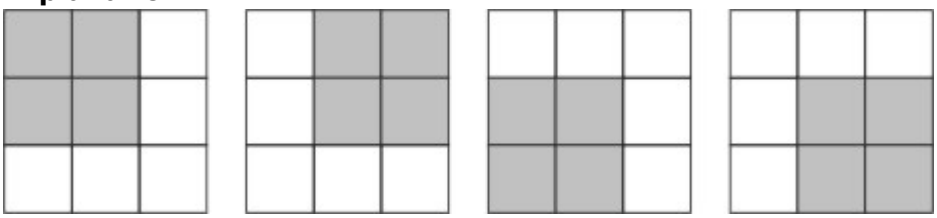


How many figures of the type as shown above are there in the figure below?



A 1 **B** 2 **C** 3 **D** 4

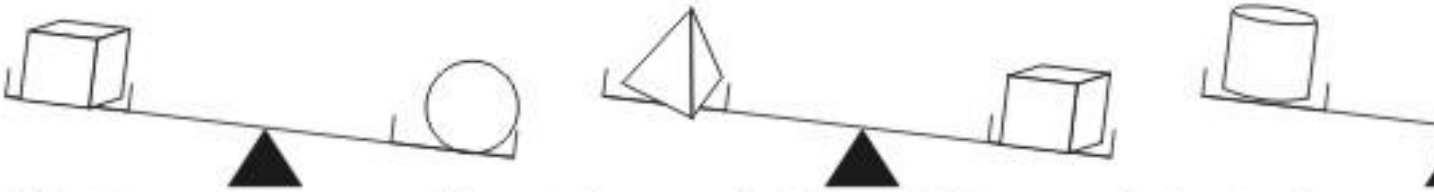
Explanation:



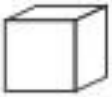
Q 3

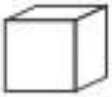
, , and are four different objects.

Weighings done with these objects are shown here:



What can we say about the weights of these objects?

A is heavier than  .

B is heavier than  .

C is heavier than .

D is heavier than .

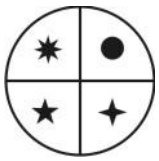
Explanation:

D :

It can be seen that and are lighter than the whereas is heavier than . So, B cannot be the answer. Also and any of the remaining objects cannot be of the same weight. So A and D cannot be the answer. The objects in C could weigh the same.

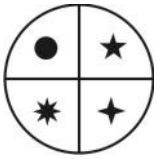
Q 4 Look at the object:

Which of the following is the same object as the one shown above? (One of them is the same object, which has just been rotated.)

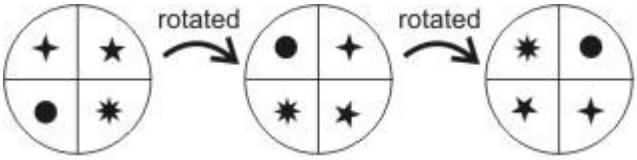
A


B

C

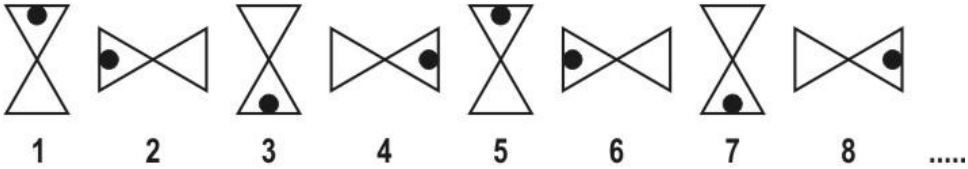
D


Explanation:

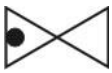


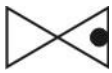
is rotated as shown in the figure.

Q 5 Observe the figures in the following pattern.



If the above pattern is continued, which of the following figures would come in the 54 th place ?

A


B


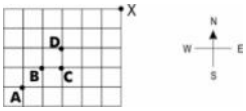
C

D

Explanation:

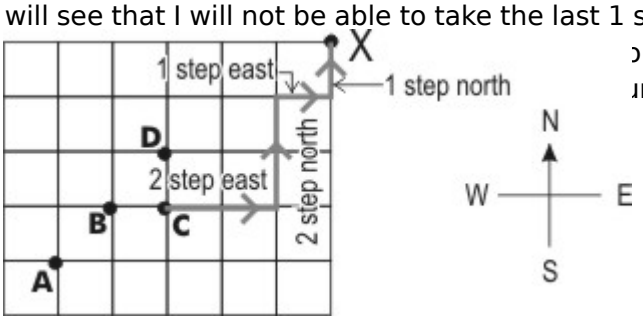
A: We see that each figure is got from the previous figure by rotating it by a quarter turn, anticlockwise. 4 quarter turns make a full turn, so after every 4 figures, we come back to the first figure and then the pattern repeats. See that the 5 th figure is same as the 1 st , 6 th is the same as the 2 nd , 7 th is the same as the 3 rd , 8 th is the same as the 4 th and so on. With the 52 nd figure we would complete 13 full rotations (13 × 4 = 52) and the 53 rd figure will be the same as the first figure and the 54 th will be the same as the second one. We see that option A is the same as the second figure and is hence the answer.

Q 6 In the grid below, I took 2 steps east, 2 steps north, one step east and 1 step north and reached X. Where did I start?



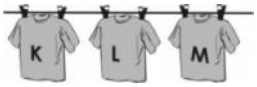
Explanation:

C: I am moving on a grid line and one side of a square is my one step. If I were to start from D, you will see that I will not be able to take the last 1 step north. If I start from B, I will reach the point to



ould reach if I start from A? Am I reaching X? Only if ure.

Q 7 The three T-shirts hanging on the line below belong to Keya, Lata and Meera - but not in that order. The letter on a T-shirt does NOT match the first letter of the name of the girl who owns it.



Also, the T-shirt hanging in the middle is Meera's.
Which T-shirt belongs to whom?

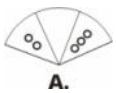
A K- Keya's, L - Lata's, M- Meera's
B K- Lata's, L - Meera's, M- Keya's
C K- Meera's, L - Keya's, M- Lata's
D K- Keya's, L - Meera's, M- Lata's

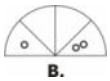
Explanation:

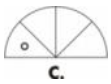
B: We are give that the T-shirt hanging in the middle is Meera's. So, of the other 2 T-shirts, with the letters K and M on them, belong to Lata and Keya. The T-shirt with K on it cannot belong to Keya, as then the letter on the T-shirt would match the first letter of her name. So it belongs to Lata. So the T-shirt with the letter M belongs to Keya.

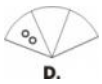
Q 8 See the paper wheel shown below:

Which of the pieces below is a part cut out from this paper wheel?

A 
A.

B 
B.

C 
C.

D 
D.

Explanation:

D: In the given figure, there are 2 empty parts between the ones with 2 circles and 3 circles, whereas in option A, there is only 1. So A cannot be the answer. Similarly in the given figure there is only one empty part between the ones with 1 circle and 2 circles, where as there are 2 empty parts between these 2 in option B. So B also cannot be the answer. C cannot be the answer too, as we don't have 3 continuous empty parts in the given figure. Comparing D with the given figure we see that it matches with a part of the given figure.

Skill: Arithmetic operations: four basic operations, properties (61%)

Q 1 If a number is divided by 9, what is the greatest remainder it can leave?

A 10 **B** 9 **C** 8 **D** 1

Explanation:

C: The remainder when a number is divided by 9 is what remains when we have taken away as many 9s as possible from the number. So the remainder has to be less than 9. If we have a remainder 9 or greater, it means that we can still take away 9. So the greatest remainder possible when a number is divided by 9 is 8.

Q 2 In her rough work, Janu added 2389 and 740 as shown here.

Is it okay to write the number 740 this way in the addition?

(*Hint: You do NOT need to calculate*)

A Yes, because 0 has no value so 74 is the same as 740.
B No, because the answer will be wrong if 0 is not written.
C No, because the digits 7 and 4 have NOT been arranged according to their value.
D Yes, because 7 is added with the hundred's digit and 4 with the ten's digit.

Explanation:

D: 740 is 7 hundreds and 4 tens. So adding 740 to 2389 is the same as adding 7 hundreds and 4 tens. Janu has added 7 to the hundred's digit in 2389 and 4 to the ten's digit, and so she is correct. She has to add 0 to the ones digit of 2389, as there are 0 ones in 740. So it is alright even if she doesn't write the 0, as long as she is adding the *7 to the hundred's digit and 4 to the tens digit* . Also 74 is NOT the same as 740, just as 1 is not the same as 10 or even 100.

Q 3 Balbir, Gunjan and Alya have written 3 number sentences.

Balbir:	**815 = 800 + 15**
Gunjan:	**800 + 15 = 815**
Alya:	**800 + 10 + 5 = 815**

Whose number sentence is correct?

A only Gunjan's
B only Alya's
C only Gunjan's and Alya's
D All of them are correct.

Explanation:

D: The equal sign tells us that the expressions on both its sides have the same value. Here the expressions on both sides of the = sign in all the 3 number sentences have the same value, and so all the three sentences are correct. "=" does not mean that what comes to its right has to be the answer to the problem on its left. When we write 12 = 7 + 5, we are just saying that 12 is equal in value to 7 + 5.

Q 4 An airline allows 20 kg baggage for each person. Any extra weight above that is charged at Rs. 70 per kg.

i.e. if one carries 1 kg extra, one will have to pay Rs. 70.

Anita's luggage weighed 25 kg. What amount will she have to pay for the extra luggage that she carried?

A Rs. 70
B Rs. 350
C Rs. 1400
D Rs. 1750

Explanation:

B: Anita does not have to pay anything for the first 20 kg of her luggage. She has to pay only for the extra luggage that she carries over 20 kg. She carries 25 - 20 = 5 kg extra luggage. For each kilogram of extra luggage, she has to pay Rs. 70. So for 5 kg she has to pay 5 × 70 = Rs. 350.

Q 5 □

÷ 100 = 50
The number in the empty box is

A 2
B 150
C 500
D 5000

Explanation:

D: Here we need to find out what number when divided by 100, will give an answer 50. Clearly 100 × 50 = 5000 gives an answer 50 when divided by 100. Note that the question is NOT asking for 100 ÷ 50.

Q 6 The sum of 2 numbers is 250. One of them is greater than 150 .

Which of these is definitely true about the other number?

A It is equal to 100.
B It has to be less than 100.
C It has to be greater than 100.
D It has to be a number between 150 and 250.

Explanation:

B: The sum of 2 numbers is 250- this is the same as saying that if you have some marbles, and I have some number of marbles, the total number of marbles is 250. Now if you have 150 marbles and the total number is 250, how many will I have? 250-150 or 100. So if you have more than 150 marbles and the total number is 250, how many will I have? It has to be less than 100, because if I had more, then the total would be more than 250! So the answer is B.

Q 7 76 × 2 = 152 and 76 × 8 = 608. Which of these is equal to 76 × 28?

(Note: You need not actually multiply or add.)

A 152 + 608 **B** 1520 + 608 **C** 6080 + 152 **D** 152 × 608

Explanation:

B: 76 × 28 is 28 times 76, that is 20 times 76 + 8 times 76. We are given that 76 × 2 = 152, so 76 × 20 is 1520. We are also given 76 × 8 = 608. So 20 times 76 + 8 times 76 = 1520 + 608.

Q 8 8000 ÷ 75 gives quotient 106 and remainder 50.

If 8000 is divided by 74, the quotient must be

A greater than 106 **B** less than 106 **C** exactly 106 **D** less than 50

Explanation:

A: The given statement means that if we divide 8000 chocolates equally among 75 people, each gets 106 chocolates and 50 chocolates remain. Now dividing 8000 by 74 is equivalent to dividing 8000 chocolates among 74 people. We are dividing the same number of chocolates, among *fewer* people, so each would get *more* number of chocolates. So the quotient would be greater than 106. So we can say that when the dividend is the same, the smaller the divisor and larger the quotient.

Notes